Assume/Guarantee Contract of `ThreeAxisController`

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1 Variable Declaration

Maps : `variables` in Code $\mapsto$ variables in Formal Specification

Alphabet :
workmode, pAngle, pRate, u′, destRate, destRate′, mController, mController′, CTRL_PARAM_SAM, CTRL_PARAM_SAM′, clock, clock′

Inputs :

- `workmode` $\mapsto$ workmode : $\mathbb{N}$
- `pAngle` $\mapsto$ pAngle : $\mathbb{N} \rightharpoonup \mathbb{R}$
- `pRate` $\mapsto$ pRate : $\mathbb{N} \rightharpoonup \mathbb{R}$

Outputs :

- `u` $\mapsto$ u : $\mathbb{N} \rightharpoonup \mathbb{R}$

In-Outputs : None

States :

- `destRate` $\mapsto$ destRate : $\mathbb{N} \rightharpoonup \mathbb{R}$
- `mController` $\mapsto$ mController : $\mathbb{N} \rightharpoonup \{\texttt{Up} : \mathbb{R}, \texttt{Ud} : \mathbb{R}, \texttt{fy} : \mathbb{R}\}$

Parameters :

- `CTRL_PARAM_SAM` $\mapsto$ CTRL_PARAM_SAM : $\mathbb{N} \rightharpoonup \{\texttt{Kp} : \mathbb{R}, \texttt{Kd} : \mathbb{R}\}$

Clocks : clock, clock′ : $\mathbb{R}_{\geq 0}$

2 AG Contract

Assumptions :

$$\begin{aligned} Assumptions \triangleq\ & \mathsf{len}(\mathsf{pAngle}) \geq 2 \\ & \wedge\ \mathsf{len}(\mathsf{pRate}) \geq 3 \\ & \wedge\ \mathsf{len}(\mathsf{destRate}) = \mathsf{len}(\mathsf{mController}) = \mathsf{len}(\mathsf{CTRL_PARAM_SAM}) = 3 \end{aligned}$$

Guarantees :

$$Guarantees \triangleq Check_1 \wedge Check_2 \wedge Check_3 \wedge Check_4 \wedge Check_5 \wedge Check_6$$

$$\begin{aligned} Check_1 \triangleq\ & \mathsf{len}(\mathsf{u}') = \mathsf{len}(\mathsf{mController}') = 3 \\ & \wedge\ \mathsf{CTRL_PARAM_SAM}' = \mathsf{CTRL_PARAM_SAM} \\ & \wedge\ \mathsf{clock}' = \mathsf{clock} \end{aligned}$$

$$\begin{aligned} Check_2 \triangleq\ & \textbf{if } \mathsf{workmode} = \texttt{0x00} \ \vee\ \mathsf{workmode} = \texttt{0x33} \textbf{ then} \\ & \quad \mathsf{destRate}' = \{0 \mapsto 0,\ 1 \mapsto 0,\ 2 \mapsto \mathsf{destRate}[2]\} \\ & \textbf{elif } \mathsf{workmode} = \texttt{0x11} \textbf{ then} \\ & \quad \mathsf{destRate}' = \{0 \mapsto 0,\ 1 \mapsto -0.5,\ 2 \mapsto \mathsf{destRate}[2]\} \\ & \textbf{elif } \mathsf{workmode} = \texttt{0x22} \textbf{ then} \\ & \quad \mathsf{destRate}' = \{0 \mapsto 0.5,\ 1 \mapsto 0,\ 2 \mapsto \mathsf{destRate}[2]\} \\ & \textbf{else} \\ & \quad \mathsf{destRate}' = \mathsf{destRate} \\ & \textbf{endif} \end{aligned}$$

$$Check_3 \triangleq \mathsf{mController}'[0] = \left\{ \begin{array}{lll} \texttt{Up} & \mapsto & \max\{-8, \min\{\mathsf{pAngle}[0], 8\}\}, \\ \texttt{Ud} & \mapsto & \max\{-1.2, \min\{\mathsf{pRate}[0] - \mathsf{destRate}'[0], 1.2\}\}, \\ \texttt{fy} & \mapsto & \mathsf{mController}'[0].\texttt{Up} \times \mathsf{CTRL_PARAM_SAM}[0].\texttt{Kp} \\ & & + \mathsf{mController}'[0].\texttt{Ud} \times \mathsf{CTRL_PARAM_SAM}[0].\texttt{Kd} \end{array} \right\}$$

$$Check_4 \triangleq \mathsf{mController}'[1] = \left\{ \begin{array}{lll} \texttt{Up} & \mapsto & \max\{-8, \min\{\mathsf{pAngle}[1], 8\}\}, \\ \texttt{Ud} & \mapsto & \max\{-1.2, \min\{\mathsf{pRate}[0] - \mathsf{destRate}'[1], 1.2\}\}, \\ \texttt{fy} & \mapsto & \mathsf{mController}'[1].\texttt{Up} \times \mathsf{CTRL_PARAM_SAM}[1].\texttt{Kp} \\ & & + \mathsf{mController}'[1].\texttt{Ud} \times \mathsf{CTRL_PARAM_SAM}[1].\texttt{Kd} \end{array} \right\}$$

$$Check_5 \triangleq \mathsf{mController}'[2] = \left\{ \begin{array}{lll} \texttt{Up} & \mapsto & 0, \\ \texttt{Ud} & \mapsto & \max\{-1.2, \min\{\mathsf{pRate}[2], 1.2\}\}, \\ \texttt{fy} & \mapsto & \mathsf{mController}'[2].\texttt{Up} \times \mathsf{CTRL_PARAM_SAM}[2].\texttt{Kp} \end{array} \right\}$$

$$Check_6 \triangleq \forall 0 \leq i < 3 \cdot \left(\mathsf{u}'[i] = \max\{-1.3,\ \min\{\mathsf{mController}'[i].\texttt{fy}, 1.3\}\} \right)$$